

Introduction

1 in 7 people report using AI chatbots instead of contacting a GP
1 in 5 decided against professional advice on that basis

Sport & Exercise Medicine (SEM) is unusually exposed because its interventions span weeks/months without direct clinician supervision. This review investigated the use of AI in SEM and adjacent fields to:

- 1) Appraise the evidence base and regulatory landscape;
- 2) Evaluate opportunities and risks;
- 3) Deliver priorities for responsible development, validation & implementation.

Methods

Search strategy

A structured literature search and analysis was conducted using *Aletheia* and *Oghma* – search, evaluation, and analytic engines built on a bespoke research platform under development by the group (Symposion, Aurora Performance, UK). This included:

- Conventional biomedical science databases
- Focused search of major organisations
- Citation searching & snowballing

Included literature

Peer-reviewed articles, grey literature, frameworks & standards, legal & regulatory documents, real-world cases.

Study webpage

Results

139 sources from across the domains of:

- Biomedicine: SEM, Orthopaedics, Allied MSK
- Infomatics: Digital health, software engineering, AI
- Regulatory: Legal & governance standards

Four domains of realistic application to SEM:

- Treatment: Injury management & rehabilitation
- Prevention: Injury & illness prevention
- Decision support: Optimising clinical management
- Health data science research, learning, 'meta-audit'

Classification of AI tools according to end user:

- Provider: clinics, services, clinicians, organisations
- Service Users: patients, athletes (supervised / unsupervised)
- Non-clinical: researchers, developers, administrators

Poor evidence: all contexts, especially in athletes, sport, nutrition
AI Chatbots: widespread use (1 in 7 public), problematic in ~50%
Acceptability: evidence of engagement decline if unsupervised
Governance: mostly voluntary, little unified regulatory oversight
Consequence: higher in unsupervised or decision-support tools
Priorities: validation, stakeholder-led design, reporting standards

Applications, evidence and governance

Artificial intelligence in Sport, Exercise & Musculoskeletal Medicine (SEM): what the evidence shows across four domains of use

